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Fine-grained, aspect-based sentiment analysis on economic and financial lexicon

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ABSTRACT

Extracting sentiment from news text, social media and blogs has recently gained increasing interest in economics and finance. Despite many successful applications of sentiment analysis (SA) in these domains, the range of semantic techniques employed is still limited and predominantly focused on the detection of sentiment at a coarse-grained level. This paper proposes a novel methodology for Fine-Grained Aspect-based Sentiment (FiGAS) analysis. The aim is to identify the sentiment associated with specific topics of interest in each sentence of a document and to assign real-valued polarity scores between -1 and +1 to those topics. The proposed approach is unsupervised and customised to the economic and financial domains by using a specialised lexicon provided by us along with the FiGAS source code. Our lexicon-based SA approach relies on a detailed set of semantic polarity rules that allow understanding of the origin of sentiment – in the spirit of the recent trend on *Interpretable AI*. We provide an in-depth comparison of the performance of the FiGAS algorithm with other popular lexicon-based SA approaches in predicting a humanly annotated dataset in the economic and financial domains. Our results indicate that FiGAS statistically outperforms the other methods by providing a sentiment score that is closer to those of human annotators.

1. Introduction

There is increasing interest in economics and finance in the application of sentiment analysis (SA) to textual data [1–4]. SA is a technology directly related to natural language processing (NLP), aiming at understanding and measuring the emotional content and polarity of a textual message, such as a document, interaction, or event [5–7]. Its outcome may be quantitative/qualitative polarity (e.g., $[-1, +1]$, *extr. neg.*, *neg.*, *neut.*, *pos.*, *extr. pos.*, etc.) or an emotional state (e.g., *joy*, *anger*, etc.). The goal is to extract relevant information from textual sources such as financial reports, economic press releases, news articles and social media. This information can be used to enrich and sharpen economic analysis [8], as in the case of the application of sentiment analysis to forecast economic and financial variables. In particular, sentiment derived from news is especially useful when forecasting macroeconomic variables, since it allows the state of the economy to be monitored in real-time, as opposed to the official releases of macroeconomic variables that are seldom available and with significant delays [4,9,10].

Several studies use sentiment from textual data (e.g., social media, newspapers, or economic and financial microblogs) to forecast financial variables [1,11–14]. As the famous American financier Bernard Baruch stated long ago, *"... what actually registers in the stock market's fluctuations are not the events themselves, but the human reactions to these events"* (taken from [15]). This statement suggests that SA could greatly improve forecasting models [16,17] and provide more accurate forecasts that can serve as the basis for more informed economic and financial decisions [18]. A major source of textual data used in economic and financial SA is represented by newspaper articles. These articles are particularly informative because they discuss key events, economic press releases and expert opinions that affect consumers' perception of the economy. This is because consumers interpret the tone and volume of economic reporting as informative of the state of the economy. In particular, higher volumes of economic news increase the likelihood that consumers will adapt their expectations of current conditions in comparison to periods of low coverage [19].

SA applications in economics and finance are based on either lexicon-based (e.g., [11,12,20,21]) or machine learning methods (e.g., [15,22–24]). A lexicon-based SA approach [25] represents an unsupervised technique that builds on a dictionary of words that are assigned positive or negative sentiment values. The words in a sentence are associated with sentiment polarity scores, which

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are then aggregated at sentence level using a combining function, such as the word count, sum or average [5]. One primary issue with this approach is that it requires a sentiment dictionary that is customised to reflect the domain lexicon specific to the application. As discussed in detail in Section 2, general purpose dictionaries may not be comprehensive and accurate enough for specific domains. Furthermore, most lexicon-based SA methods focus on a coarse-grained analysis – i.e., either positive or negative scores – of the sentiment expressed in the text (see [4,23,26,27], among other works). However, coarse-grained methods may not be sufficiently accurate in evaluating the polarity of an economic or financial topic since the sentiment of the entire text may not directly relate to that specific topic [15]. As an example, consider the following sentence: "*Italian GDP continues to grow, despite industrial production shrinking and the agricultural sector suffering from the recent adverse meteorological conditions that hit the country*". From the perspective of the *Italian GDP*, the sentiment is clearly positive. However, the sentiment expressed for *industrial production* and the *agricultural sector* is most likely negative. In this case, a coarse-grained SA approach would assign a negative sentiment to the entire sentence since the majority of terms in the text have a negative connotation. This could lead to a very inaccurate representation of the overall sentiment if the researcher is interested in extracting the tone regarding a particular economic topic.

An alternative approach consists of a fine-grained textual analysis that concentrates on a specific term of interest using supervised machine learning models [15,24,28,29]. Some recent applications in economics and finance include recent work on technical analysis [30,31], sentiment word embeddings [32,33], attention-based deep learning models [34,35], and forecasting sentiment intensity using stacked ensemble methods [22,36,37]. One key disadvantage of such supervised approaches consists of their dependence on labelled data: it is indeed extremely difficult to obtain sufficient and correctly labelled data for a specific domain. In addition, the transparency of a lexicon-based approach represents a significant advantage in fields such as economics and finance, where stakeholders favour the clear interpretability of the methods and outcomes.

In this paper we propose a novel lexicon-based SA algorithm developed specifically for the economic and financial domains that we term Fine-Grained Aspect-based Sentiment (FiGAS). The goal of FiGAS is to compute the score of a sentence that reflects the sentiment regarding a specific topic of interest, rather than the overall sentiment of the sentence. The resulting score is a floating point value in the range between -1 (extremely negative/bearish) and $+1$ (extremely positive/bullish), with 0 representing a neutral sentiment. The FiGAS sentiment score has two characteristics [38]. Firstly, it is based on a fine-grained dictionary that has been specifically developed for applications in the economic and financial domain. Secondly, the sentiment is calculated solely on the terms in the sentence that are semantically connected to the topic of interest based on a set of linguistic rules. While lexicon-based methods with these characteristics are available for general text [39], the innovation of this paper consists of proposing a lexicon-based fine-grained SA method that is specifically tailored to the economic and financial domain. The choice of a lexicon-based fine-grained SA approach – compared to a machine-learning approach – produces a completely unsupervised algorithm that does not require collecting and labelling a training corpus. In addition, the FiGAS algorithm is *interpretable* [40,41] since the sentiment score can be transparently explained based on the semantic rules and the polarity score for each word.

FiGAS can be categorised as a SA approach from *sentic computing* [42], which aims at improving algorithmic performance by including semantic features in opinion mining [1,43]. In other words, the goal is to shift from a word-level to a concept-level analysis of sentiment. For a survey on sentic computing and discussions on new trends in SA, we refer to [44]. We test the performance of FiGAS in detecting positive and negative sentiment of a specific topic by comparing it with other popular lexicon-based SA approaches. The evaluation is based on a set of economic and financial sentences labelled by professional human annotators. The results show that FiGAS outperforms the other methods and, in most cases, delivers sentiment scores that are similar to those of the annotators.

The remainder of the paper is organised as follows. Section 2 describes the lexicon-based SA approach and the dictionaries that are typically used in general or specific applications. The technical details of our FiGAS algorithm are provided in Section 3. The results of the empirical evaluation of the algorithm are discussed in Section 4. In this section we also examine the performance of the algorithm compared to other popular lexicon-based SA approaches on a humanly annotated dataset in the economic and financial domain (Section 4.1). In Section 4.2, we discuss the construction of economic sentiment indicators using FiGAS on a large news corpus, with potential forecasting and nowcasting of economic and financial variables. Finally, in Section 5 we summarise the results and discuss some future directions to extend this work.

2. Lexicon-based SA and dictionaries

Lexicon-based SA methods measure semantic sentiment or sentiment polarity by aggregating the sentiment of the individual words or phrases [25]. The key element in the analysis is a dictionary that assigns each word in the lexicon a sentiment score. For instance, the terms *wonderful* and *growing* are examples of words expressing a positive sentiment, whereas the terms *disgusting* and *bad* convey a negative sentiment. More generally speaking, the sentiment dictionary can assign a real-valued number to each word, ranging between a minimum and a maximum sentiment score (e.g., between -1 and $+1$, or between 0 and 100 in most cases). Alternatively, dictionaries may focus on the classification of words into emotion categories, such as *anger*, *joy* or *fear*.

In a lexicon-based SA method, a text is represented using a bag-of-words model [45], which accounts for word frequencies but disregards other information such as grammatical structure or word order. The dictionary is then used to characterise each word in the text with a sentiment, these sentiments then being aggregated to compute an overall sentiment score for the text. When using a dictionary where terms are assigned a sentiment polarity score, the most common choice for calculating the sentiment of the text is to compute the average polarity of the individual words [1,2,5,14,39], that is:

$$\sum_{i=1}^{k} s(w_i)/k, \tag{1}$$

where k is the total number of words in the text that are included in the lexicon, and $s(w_i)$ is the sentiment polarity score associated with the word w_i. Alternatively, if the lexicon only lists positive, negative and neutral terms, a sentiment polarity value for the entire text is obtained by counting the words in each category [4,21,46], that is:

$$(p-n)/k, \tag{2}$$

where k was defined earlier, and p and n are the total number of positive and negative words in the text, respectively.

Most of the existing SA approaches evaluate the sentiment polarity of a certain text by considering all of the terms that convey sentiment in the text. This approach is suitable for short documents, such as tweets or news headlines, since it can be assumed that the majority of terms in the text contribute to the overall sentiment. However, longer documents, such as news articles, typically discuss several topics of interest: these methods may deliver an inaccurate measure of sentiment since they aggregate the polarity of all confounded topics. To overcome this limitation, several papers have recently proposed fine-grained SA methods [15,24,39]: their novelty is that they focus on the parts of the text that express sentiment with respect to a specific topic of interest, and analyse these expressions in a fine-grained manner. Our proposed SA method belongs to this category, as we describe in detail in Section 3.

The most important element of these methods is the sentiment lexicon. A SA lexicon can be created through human annotation or by using an automatic generation mechanism [5]. Lexicons can differentiate between being general-purpose (i.e., used for various applications since they are constructed to cover a broad and common lexicon), or domain-specific (i.e., specialised for a specific application domain). Another dimension to consider is the language of the lexicon itself. The majority of sentiment lexicons have been constructed for the English language, while working with other less popular or several languages remains an ongoing challenge in SA [1,6,25]. Should the sentiment lexicon be unavailable for the language of interest, the text can be translated [2]. However, it is typically the case that such translation results in a poorer performance of SA methods.

In the rest of the section, we describe the sentiment lexicons that we consider in our empirical analysis. They are popular sentiment dictionaries for the English language, ranging from general lexicons to domain-specific ones, and from simple word lists to polarity scores. In Section 2.5, we describe the dictionary that we have constructed custom for the economic and financial domain.

2.1. SentiWordNet

SentiWordNet [47] is a general-purpose dictionary where each term included in the lexicon has a polarity score in the $[-1, +1]$ range and is linked to its synsets (i.e., its set of synonyms derived from the WordNet database [48]). WordNet[1] is an open-source lexical resource for the English language that is extensively used in NLP applications. It categorises nouns, verbs, adjectives and adverbs into groups of cognitive synonyms (i.e., synsets), with each synset being associated with a specific concept and linked to others via lexical and conceptual–semantic relations.

Term relationships contained in WordNet create a highly disconnected graph [49]. Therefore, expanding opinion information, such as in SentiWordNet, needs to be bound only to a subset of seed terms. To overcome this, the term's sentiment can be inferred by examining the information contained in the term's glossaries (i.e., explanatory text linked to each term), under the assumption that the term and its glossaries have similar polarity. Leveraging on this assumption, a method for lexicon expansion is proposed in [50], which assigns positive or negative opinions based on the existence of words in the glossary that are known to carry sentiment. The assumption in support of this method is that glossaries have a potentially low level of noise since (i) they are designed to match the term meaning as closely as possible and (ii) they have a relatively standard style, grammar and syntactic structure [50].

A two-stage lexicon expansion approach was also used to construct SentiWordNet [47]. Firstly, WordNet term relationships such as synonyms, antonyms and hyponyms are explored to develop a core of seed words known a priori to carry positive or negative sentiment bias (see [47] for more details). After a fixed number of iterations, a subset of WordNet terms is obtained with either a positive or negative label. These terms' glossaries are then used to train an ensemble of machine learning classifiers. To minimise bias, the classifiers are trained using different set sizes. The predictions from this ensemble of classifiers are then used to determine the sentiment of the remainder of the terms in WordNet. At the time of writing, SentiWordNet contains 28,431 sentiment terms, out of total 86,994 WordNet terms.[2]

2.2. SenticNet

SenticNet [51] is a general-purpose lexical resource for concept-level sentiment analysis, whose main aim is to make the conceptual information conveyed by natural language more easily accessible and interpretable by machines. SenticNet associates word polarity with more complex concepts, by replacing the classic bag-of-words model [45] with a new language model that sees text as a bag of concepts and narratives. This approach goes beyond counting word co-occurrence frequencies to classify text by leveraging an ensemble of symbolic and sub-symbolic AI techniques to gain a deeper understanding of natural language.

The different versions of SenticNet are accessible in standard machine-readable format (i.e., RDF/XML/OWL).[3] SenticNet 1 simply associates polarity scores to almost 6 thousand concepts. In addition to polarity, SenticNet 2 also assigns semantics to common-sense concepts and extends the scope of the knowledge base to around 13,000 entries. SenticNet 3 broadens the spectrum of the semantic network to approximately 30,000 concepts, while SenticNet 4 introduces the concept of semantic primitives to further extend the knowledge base to 50,000 entries. SenticNet 5 reaches 100,000 common-sense concepts by employing recurrent neural networks to infer primitives by lexical substitution.

The currently available version is SenticNet 6 [53], which provides sentiment scores (in a range between -1 and 1) for around 200,000 common-sense concepts by leveraging both symbolic models (e.g., logic and semantic networks) to encode meaning, and sub-symbolic methods (e.g., BiLSTM and BERT) to infer syntactic relations (more details are provided in [53]). The latest version of the knowledge base can also be browsed as an ontology and selectively accessed online through dedicated REST APIs – available in 40 different languages including English.

The sentiment conveyed by each term in SenticNet is defined based on the intensity of 24 basic emotions as proposed in the Hourglass of Emotions model from [52] (see Fig. 1). The model can be used to measure how much an individual is (i) self-aware of his/her own ideas, thoughts, and feelings (*Introspection*); (ii) prone to get angry quickly (*Temper*); (iii) confident in interaction benefits (*Aptitude*); and, finally, (iv) comfortable with interaction dynamics (*Sensitivity*). In particular, SenticNet describes each affective dimension with six activation levels, which measure the strength of the related expressed/perceived emotion. The combination of the six levels and the four affective dimensions produces the 24 basic emotions used by the model illustrated in Fig. 1: the vertical dimension represents the intensity of each affective dimension, while the radial dimension reflects the different emotional configurations.

The overall sentiment polarity of a concept is strongly connected to its attitude and feeling, and can be derived from the four

[1] WordNet, A Lexical Database for English. Available at: https://wordnet.princeton.edu/.

[2] SentiWordNet, version 3.0, available at: https://github.com/aesuli/SentiWordNet.

[3] SenticNet, available at: https://sentic.net/.

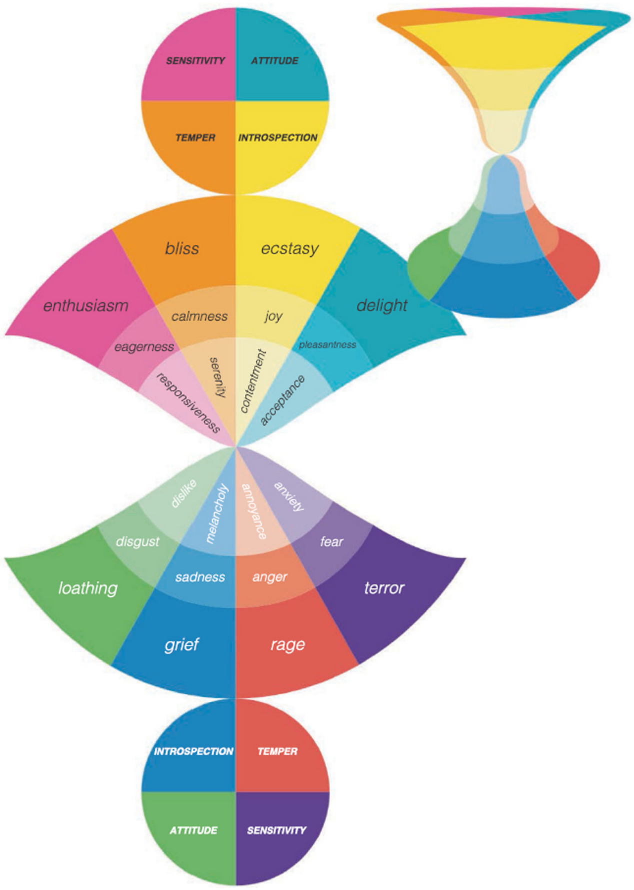


Fig. 1. Illustration of the Hourglass of Emotions model [52]. Since affective states are represented according to their strength (from strongly positive to null to strongly negative), the model assumes an hourglass shape.

affective dimensions of the Hourglass model using the following expression [52]:

$$s_c = \frac{I_c + T_c + A_c + S_c}{|\operatorname{sgn}(I_c)| + |\operatorname{sgn}(T_c)| + |\operatorname{sgn}(A_c)| + |\operatorname{sgn}(S_c)|}, \tag{3}$$

where c is an input concept, s_c represents the sentiment polarity score, I is the value of *Introspection* (providing a *joy*-versus-*sadness* measure), T is the value of *Temper* (providing a *calmness*-versus-*anger* measure), A is the value of *Attitude* (providing a *pleasantness*-versus-*disgust* measure), and S is the value of *Sensitivity* (providing a *eagerness*-versus-*fear* measure).

In Eq. (3), the denominator will actually be equal to 1 for most of the cases, since the majority of concepts are only associated with one emotion; it will be equal to 2 for concepts that are associated with bidimensional emotions like, e.g., *love* (*joy*+*pleasantness*) and *submission* (*fear*+*pleasantness*); it will be equal to 3 for those few concepts that are associated with tridimensional emotions like, e.g., *bittersweetness* (*sadness*+*anger*+*pleasantness*); finally, it will be 4 for those very rare concepts that are associated with compound emotions that span all dimensions like, e.g., *jealousy* (*anger*+*fear*+ *sadness*+*disgust*) [52].

2.3. Harvard IV-4 and general inquirer

The General Inquirer (GI)[4] is one of the first and most influential tools in the content analysis of textual data [54]. GI classifies English terms into 182 categories that are collected from various sources, including the *Harvard IV-4* dictionary.[5] While the Harvard IV-4 was originally developed for psychological applications, it has been used as a general-purpose lexical resource for a wide variety of other fields, such as political science, survey research and business marketing, among others. The sentiment lexicon assigns only one class to each word, rather than a fine-grained, real-valued score representing the strength of each class categorisation.

There are several applications in finance of the GI Harvard IV-4 sentiment lexicon. Tetlock [11] analyses the *Wall Street Journal* column "Abreast of the Markets" using 77 categories and links the daily news tone to stock market returns. In particular, he finds that high levels of pessimism in news are associated with lower subsequent stock returns. Later applications of the Harvard IV-4 in finance have primarily used the positive and negative categories (2291 and 1915 words, respectively, if we account for various inflections of the same words). One example is [55], which finds that only negative words included in firm-specific news help forecast stock prices.

Follow-up studies on the Harvard IV-4 dictionary criticise its usage in non-psychological applications, due to fact that the polarity of a word heavily depends on the context in which the word is used. For instance, Loughran and McDonald [56] report that approximately 75% of the negative words from the Harvard IV-4 dictionary do not have a pessimistic connotation when used in a financial context. Some examples are the words *tax*, *cost*, *capital*, *board*, *liability* and *depreciation*, which frequently appear in financial reports such as the 10-K filings. In addition, words such as *crude*, *cancer* or *mine* are classified as negative by the Harvard IV-4 lexicon, despite their potential relation to discussions of the oil, pharmaceutical or mining industries, respectively. The conclusion of [56] is that using a word classification scheme developed in domains other than the application at hand may lead to misclassification and spurious correlations. As we will discuss in Section 4, our results also confirm these findings.

2.4. Loughran–McDonald

The Loughran and McDonald (LMD) lexicon [56,57] was developed to fill the gap regarding a lexicon specific to the financial domain in the English language. LMD includes approximately 4000 words, categorised as follows: negative, positive, uncertainty, litigious, constraining, strong modal and weak modal.[6] As also applies in the case of the LMD lexicon, words are simply classified into categories and are not assigned a fine-grained score. Loughran and McDonald [56] constructed the LMD dictionary by analysing word usage in a large sample of 10-K financial statements between 1994 and 2008. This allowed them to create different lists based on word frequencies. They also show that the tone computed from their negative lexicon has predictive power for filing period excess stock returns and that it performs better relative to the tone based on the generic Harvard IV-4 dictionary negative lexicon.

The LMD dictionary is one of the first resources customised for sentiment analysis in a business context and has been used in many applications to the financial and economic domains. For instance, Garcia [26] uses the LMD positive and negative word lists to analyse two financial columns of the *New York Times*, finding that sentiment helps predict future stock returns – during recessionary periods, in particular. On the other hand, the LMD dictionary is not exempt from shortcomings. One relevant

4 GI official web portal: http://www.wjh.harvard.edu/~inquirer/.

5 The details of the latest version of the Harvard IV-4 dictionary are available at: http://www.wjh.harvard.edu/~inquirer/homecat.htm.

6 The LMD word lists are available at https://sraf.nd.edul.

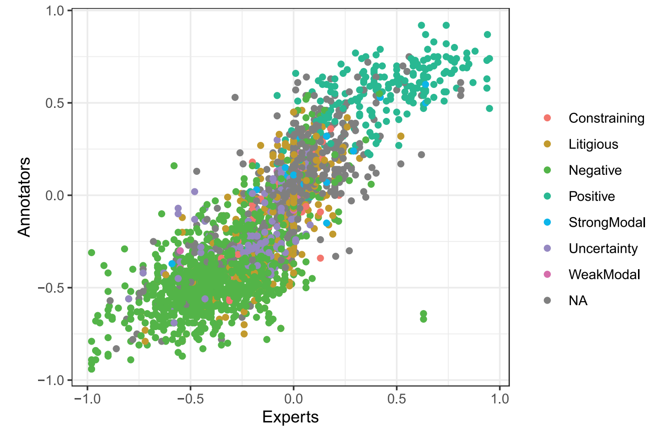
issue with LMD is that word distribution is highly unbalanced across categories, since more than half of the terms fall under the negative category and only 354 words are positive. Such an imbalance between positive and negative entries may make it difficult to use the lexicon in other domains, or even subdomains of economics and finance. As stated by Loughran and McDonald [57], the LMD dictionary was specifically developed for the analysis of 10-K financial statements, although it is now widely used in several economic and financial applications. As stressed by its proponents, the LMD lexicon should be used with caution when dealing with tasks other than those within its original scope, or else it would face the same shortcomings as the Harvard IV-4 dictionary, as discussed in Section 2.3.

2.5. SentiBigNomics

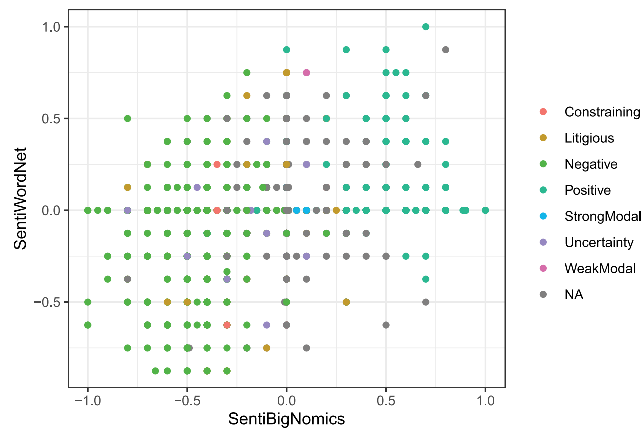
The goal of the *SentiBigNomics* dictionary is to provide users with a fine-grained sentiment lexicon for the economic and financial domains (see, for example, [2,58]). Such a dictionary is not currently available since SentiWordNet (Section 2.1) and SenticNet (Section 2.2) discussed above are fine-grained and general purpose, while LMD is coarse-grained and domain-specific. In order to fill this gap, we designed a fine-grained sentiment lexicon suitable for applications in the economic and financial domains, that we refer to as *SentiBigNomics*.[7]

To construct the proposed fine-grained lexicon, we started with the LMD word list [56] (Section 2.4), which includes approximately 4000 words. After lemmatising, we removed a small fraction of 10-K-specific terms without polarity (e.g., juris, obligor, pledgor, among others) and then manually added a list of words that frequently occur in economic documents. We then labelled the selected words by employing a total of 15 human annotations using the commercial Amazon AWS SageMaker service.[8] In particular, for each term we collected 5 different annotations from field experts (i.e., researchers active in the field of economic forecasting) and 10 human annotations from US-based professional annotators. The annotation task consisted of assigning a score in the range $[-1, +1]$ to each term based on the perception of the annotator. More specifically, they were instructed that negative score values (i.e., $[-1, 0)$) would indicate that the word is most likely used with a negative connotation in an economic context, whereas positive scores (i.e, $(0, 1]$) would characterise words that are commonly used to indicate a favourable economic situation. The annotators were also given the option to assign a null score in case they believed the word had no sentiment connotation. The magnitude of the score represents the sentiment intensity of the word as perceived by the annotator: for instance, a score close to 1 indicates an extremely favourable economic situation, while a positive value close to 0 would be associated with a mildly positive connotation. In case of doubt, the annotator could consult an example of an economic sentence containing that specific term.

The upper panel of Fig. 2 compares the mean scores assigned by the US-based professional annotators and the field experts. Generally, there appears to be an overall agreement between the two annotator groups, as demonstrated by the positive correlation between average scores. Moreover, we compared the sentiment score signs with the categories from the LMD dictionary [56], which are identified by colours in Fig. 2. We aggregate the scores obtained from the domain experts and US-based annotators by taking the median value, and plot this in the lower panel of Fig. 2 against the general-purpose SentiWordNet lexicon (Section 2.1). There seems to be a weak positive correlation between the two resources, suggesting that the general-purpose resource often disagrees with the domain-specific dictionary. This confirms the relevance of using domain-specific dictionaries compared to general purpose ones. The final dictionary list is composed of 3181 terms with an associated polarity score. The resource is publicly available in GitHub.[9]



(a) Mean scores of US-based annotators versus field experts.



(b) SentiWordNet scores versus median SentiBigNomics scores.

Fig. 2. Mean scores of US-based annotators versus field expert annotations (upper panel), and SentiWordNet scores versus median scores for the proposed SentiBigNomics resource (lower panel). Colours indicate the categories identified by [56].

3. Figas: Fine-Grained Aspect-based Sentiment analysis

The current applications of SA in economics and finance typically compute sentiment at document level and use a coarse-grained dictionary (e.g., [4,23,26,27]). As discussed in Section 1, we believe that more accurate sentiment measures can be obtained by extracting the text in the document related to a specific topic of interest and using a fine-grained and domain-specific dictionary. Our FiGAS approach heads in this direction. The algorithm uses a sophisticated semantic rule scheme [38,39,59] that is able to recognise the aspect to which the sentiment refers to, while providing a fine-grained, continuous sentiment polarity

[7] The name is an acronym for the research team that developed the resource as part of the European Commission Big Data and Forecasting of Economic Developments (BigNOMICS) project (more information is available at https://ec.europa.eu/jrc/communities/en/community/big-data-and-forecasting-economic-developments).

[8] Amazon AWS SageMaker service, available at: https://aws.amazon.com/sagemaker/.

[9] https://github.com/sergioconsoli/SentiBigNomics/blob/main/python/senti_bignomics.py.

score in the $[-1, +1]$ range. In addition, the method is completely unsupervised since it is based on the SentiBigNomics lexicon (see Section 2.5), which makes the algorithm specifically customised to the analysis of economic and financial texts.

The algorithm works as follows. Suppose that we wish to extract the sentiment polarity relating to a specific economic concept, for instance *gross domestic product*. The first step is to derive the economic concepts related to the aspect of interest. This is done by means of a SPARQL query[10] against the *World Bank Group (WBG) Ontology*.[11] The WBG Ontology provides a classification taxonomy of the economic concepts used within the World Bank Group official documents. The concepts in the ontology are stored and linked in a logical hierarchy as broader and narrower terms, which may also be related across subject areas. For the example of *gross domestic product*, the related concepts obtained are[12]: *GDP; economic development; national wealth; national income accounting; economic progress; national accounting; factor price; economic regeneration; net domestic product;* etc.

In the following step, the economic concepts are identified in the input text and a rule-based procedure extracts information associated with them. The algorithm has been developed in Python, version 3.7.6, and the source code is accessible online on the authors' GitHub page.[13] The information extraction process is based on the linguistic features of the *spaCy*[14] Python library. In particular, the pipeline relies on spaCy's *en_core_web_lg* language model,[15] which represents an English multi-task Convolutional Neural Network trained on OntoNotes and with GloVe vectors trained on Common Crawl. The library also assigns word vectors, context-specific token vectors, part-of-speech (POS) tags, dependency parse and recognises named-entities. Details of the algorithm are summarised below.

Tokenisation & lemmatisation. The input text is split into meaningful segments (*tokens*) and the uninflected forms of the words (*lemmas*) in the text are taken from WordNet.

Named-entity recognition. Named-entity recognition (NER) is an information extraction procedure that identifies and labels named-entities present in unstructured text. The named-entities are allocated in predefined classes which include, among others, person names, locations, organisations, time expressions, quantities, monetary values, companies, products or percentages. NER is based on the model trained by spaCy that provides a satisfactory accuracy of around 86%. The named-entities we predominantly use in our algorithm are the locations, organisations, person names, and time expressions.

Location detection. This heuristic procedure is used to assign the location to which a sentence is referring as its most frequent named-entity location detected in the text. This step helps filtering sentences referring to a particular location of interest indicated by the user. For example, suppose that we are interested in detecting the sentiment expressed in a given text for the topic *gross domestic product* in *Italy*. In this case, all sentences that explicitly deal with the *gross domestic product* of countries other than *Italy* are discarded from the analysis.

POS tagging. Following the tokenisation step, we perform the parts-of-speech parsing and tagging of the input text using spaCy. The spaCy parser and tagger assigns a class to each term based on the *Universal POS tags*[16] categories proposed by the *Universal Dependencies (UD) framework*.[17] The UD represents an open community framework for coherent grammar annotation (parts of speech, morphological features, and syntactic dependencies) across different languages, resulting in more than 150 semantic treebanks in 90 languages. The statistical model used by spaCy to predict the part-of-speech tags gives us an extreme level of accuracy for the task (with a declared accuracy of 97.06%). Following the parsing and tagging, we loop over each part-of-speech tags and stop only when one of the economic concepts of interest is found in the text. In this way we are able to isolate the sentences and the portions of text that are of interest to our analysis.

Dependency parsing. Dependency parsing establishes the relationship between a word and the other terms in the sentence that modify its meaning. The spaCy dependency parser is very accurate with a declared accuracy of 89.76%, and also relies on the *UD framework* to classify the parsed dependencies.[18] After a concept of interest is found in the input text, we navigate over the neighbouring tokens employing a rule-based approach that uses the POS tags and parsed dependencies. Details of the rule-based approach are provided in the next section (Section 3.1). Chunks of terms related to our search concepts are constructed in this way.

Tense detection. This experimental procedure is employed to detect the tense of a verb that is related to the concept of interest. This is an optional step that can be used to focus the analysis on the text that includes one or more selected verbal tenses (i.e., *past*, *present*, *future*). This feature of the algorithm is useful when the user is interested in exploring the sentiment expressed for an economic concept in a specific verbal tense.

Negation handling. Whenever a negation term is detected in a chuck of text, the sentiment score is adjusted (i.e., the score is multiplied by -1).

3.1. Adopted linguistic rule scheme

In this section we provide the details of the semantic rule scheme used in the NLP pipeline presented above. This scheme represents the core of our algorithm and comprises a set of linguistic rules which have been derived experimentally following in-depth natural language analysis. The heuristic rules are based on both the syntax and semantics of the text and allow for explanations of how the algorithm reaches its final sentiment polarity score. This is a convenient feature since it provides transparency and interpretability to the analysis.

We now introduce the notation[19] that is later used to describe the linguistic rules:

10 W3C SPARQL Query Language for RDF: https://www.w3.org/TR/rdf-sparql-query/.

11 World Bank Group (WBG) Ontology, available at: http://vocabulary.worldbank.org/thesaurus.html.

12 Corresponding SPARQL query for the example:
SELECT ?nodeOnPath ?Label ?OtherLabel WHERE {
?subject <http://www.w3.org/2004/02/skos/core#prefLabel> *"gross domestic product"@en .*
?subject <http://www.w3.org/2004/02/skos/core#related> *?nodeOnPath .*
?nodeOnPath <http://www.w3.org/2004/02/skos/core#prefLabel> *?Label .*
OPTIONAL{?nodeOnPath <http://www.w3.org/2004/02/skos/core#altLabel> *?OtherLabel}}.*

13 FiGAS GitHub repository: https://github.com/sergioconsoli/SentiBigNomics.

14 spaCy: Industrial-Strength Natural Language Processing in Python. Available at: https://spacy.io/.

15 English model *en_core_web_lg* from spaCy, available at: https://spacy.io/models/en.

16 Universal POS tags, available at: https://universaldependencies.org/u/pos/all.html.

17 Universal Dependencies framework, available at: https://universaldependencies.org/.

18 UD Relations, available at: https://universaldependencies.org/u/dep/.

19 For consistency, we follow the spaCy notation at https://spacy.io/api/annotation.

Table 1
Main spaCy part-of-speech tags used in our algorithm.

TAG	POS	Description
CC	CONJ	Conjunction, coordinating
IN	ADP	Conjunction, subordinating or preposition
JJ	ADJ	Adjective
JJR	ADJ	Adjective, comparative
JJS	ADJ	Adjective, superlative
MD	VERB	Verb, modal auxiliary
NN	NOUN	Noun, singular or mass
NNP	PROPN	Noun, proper singular
NNPS	PROPN	Noun, proper plural
NNS	NOUN	Noun, plural
RBR	ADV	Adverb, comparative
RBS	ADV	Adverb, superlative
VB	VERB	Verb, base form
VBD	VERB	Verb, past tense
VBG	VERB	Verb, gerund or present participle
VBN	VERB	Verb, past participle
VBP	VERB	Verb, non-3rd person singular present
VBZ	VERB	Verb, 3rd person singular present

Table 2
Main spaCy dependency parsing classes used in our algorithm.

DEP	Description
acl	Clausal modifier of noun (adjectival clause)
advcl	Adverbial clause modifier
advmod	Adverbial modifier
amod	Adjectival modifier
attr	Attribute
dobj	Direct object
neg	Negation modifier
oprd	Object predicate
pcomp	Complement of preposition
pobj	Object of preposition
prep	Prepositional modifier
xcomp	Open clausal complement

- `POS`: detected part-of-speech,
- `TAG`: part-of-speech tag,
- `DEP`: parsed dependency.

Tables 1 and 2 provide the POS tags and dependency classes that we use in the linguistic rule scheme. Based on the `POS`, `DEP` and `TAG` labels defined in these tables, the algorithm only selects a chunk in a sentence if it contains a certain token-of-interest (`ToI`). In addition, the chunk is further considered for the analysis only if it relates to the tense and location of interest to the user. Overall, the algorithm takes three inputs represented by the `ToI`, tense and location.

For *tense detection*, we rely on the spaCy tense detection procedure to associate a verb (`POS` = `VERB`) to one of the following tenses:

a. past (i.e., `TAG` = `VBD` or `TAG` = `VBN`),
b. present (i.e., `TAG` = `VBP` or `TAG` = `VBZ` or `TAG` = `VBG`),
c. future (i.e., `TAG` = `MD`).

For each chunk, we calculate a score for each of the tenses above; the score is based on the number of verbs detected with that tense, weighted by the distance of the verb to the `ToI` (the closer the verb to the `ToI`, the higher the score). We assign the tense with the highest score to the whole chunk, and we continue our analysis over that chunk if the tense corresponds to that of interest to the user.

For *location detection*, we instead rely on the NER model provided by spaCy in order to identify the following entities:

- `GPE`: countries, cities, states, etc.,
- `NORP`: nationalities, or religious or political groups,
- `LOC`: non-GPE locations, mountain ranges, etc.,
- `ORG`: companies, agencies, institutions, etc.

Firstly, we search for the desired location among the above entities and assign the most relevant location to the entire document. In the following step, we find the most relevant location within each chunk and, if no entities are detected, we also assign the document location to the chunk. The FiGAS algorithm continues the analysis of the chunk only if the assigned location is among the locations of interest to the user. An additional feature of the algorithm is a list of terms to be excluded from the analysis: if at least one of these terms is present in the chunk, then the FiGAS algorithm continues to the following item.

Once the chunk has passed all of these preliminary checks, the FiGAS algorithm continues the analysis by evaluating whether the chunk satisfies any of the rules discussed below. When a matching rule is detected, FiGAS assigns a sentiment score to each of the tokens associated with the `ToI` from the SentiBigNomics lexicon. The overall sentiment of the chunk is then computed as discussed in Section 3.2. The final sentiment polarity for the whole chunk is reversed (*negation handling*) in case the chunk contains a negation or a term with a negative connotation (i.e., `DEP` = `neg`). Below, we provide more details of the grammatical and semantic rules adopted in our algorithm.

ToI associated with an adjectival modifier

In this case the `ToI` is connected by an adjectival modifier relation (`DEP` = `amod`), i.e., an adjective phrase that modifies the meaning of the `ToI`, to a term that is:

a. an adjective (`POS` = `ADJ`), in the form of:
   i. standard adjective (`TAG` = `JJ`);
   Example: ...today's strong retail sales are cheering the markets... (`ToI` = *retail sales* $\xrightarrow{\text{amod}}$ `JJ` = *strong*).
   ii. comparative adjective (`TAG` = `JJR`);
   Example: ...it will help to channel higher electricity demand in the country... (`ToI` = *electricity demand* $\xrightarrow{\text{amod}}$ `JJR` = *higher*).
   iii. superlative adjective (`TAG` = `JJS`);
   Example: ...this year we have registered the world's biggest whisky production since the last decade... (`ToI` = *whisky production* $\xrightarrow{\text{amod}}$ `JJS` = *biggest*).
b. a verb (`POS` = `VERB`);
   Example: ...it is observed so far an increasing agricultural production in Italy... (`ToI` = *agricultural production* $\xrightarrow{\text{amod}}$ `VERB` = *increasing*).

ToI associated with an adjectival clause

The `ToI` is connected via an adjectival clause (`DEP` = `acl`), i.e., a finite or non-finite clause that modifies the `ToI`, to a term that is:

a. a verb (`POS` = `VERB`);
   Example: ...the regulation was thought to push firms gaining competitive advantage in EU markets... (`ToI` = *firms* $\xrightarrow{\text{acl}}$ `VERB` = *gaining*).

ToI connected to a verb followed by an open clausal complement or an adverbial clause modifier

The `ToI` is, in this case, associated with a verb (`POS` = `VERB`), which is connected to another term by means of one of the following relations:

a. open clausal complement ($\mathtt{DEP} = \mathtt{xcomp}$), i.e., a predicative or clausal complement without its own subject;
Example: ...commercial banks reported that mortgage demand could keep dropping... ($\mathtt{ToI} =$ *mortgage demand*$\rightarrow \mathtt{VERB} =$ *keep*$\xrightarrow{\text{xcomp}} \mathtt{VERB} =$ *dropping*).
b. adverbial clause modifier ($\mathtt{DEP} = \mathtt{advcl}$), i.e., a clause that modifies a verb or another predicate (adjective, etc.) as a modifier, not as a core complement;
Example: ...company sales will reach the bottom as they struggle with the market crisis... ($\mathtt{ToI} =$ *company sales*$\rightarrow \mathtt{VERB} =$ *reach*$\xrightarrow{\text{advcl}} \mathtt{VERB} =$ *struggle*).

ToI connected to a verb followed by an adjectival complement

The ToI is associated with a verb ($\mathtt{POS} = \mathtt{VERB}$) that is followed by an adjectival complement relation ($\mathtt{DEP} = \mathtt{acomp}$); in this case, the role of the adjectival term is to provide additional information about the verb. The adjective ($\mathtt{POS} = \mathtt{ADJ}$) can be in the form of:

a. standard adjective ($\mathtt{TAG} = \mathtt{JJ}$);
Example: ...businesses will remain cautious about investment... ($\mathtt{ToI} =$ *businesses*$\rightarrow \mathtt{VERB} =$ *remain*$\xrightarrow{\text{acomp}} \mathtt{JJ} =$ *cautious*).

ToI connected to a verb followed by an object predicate

The ToI is associated with a verb ($\mathtt{POS} = \mathtt{VERB}$) that is followed by an object predicate relation ($\mathtt{DEP} = \mathtt{oprd}$). This represents a situation in which the linked verb is connected to an adjective that provides information about the object it refers to. The adjective ($\mathtt{POS} = \mathtt{ADJ}$) can be in the form of:

a. a standard adjective ($\mathtt{TAG} = \mathtt{JJ}$);
Example: ...employees qualify for 90% protection if their company is declared insolvent... ($\mathtt{ToI} =$ *company*$\rightarrow \mathtt{VERB} =$ *declared*$\xrightarrow{\text{oprd}} \mathtt{JJ} =$ *insolvent*).
b. a comparative adjective ($\mathtt{TAG} = \mathtt{JJR}$);
Example: ...last month the FED kept the rates higher than expected... ($\mathtt{ToI} =$ *FED*$\rightarrow \mathtt{VERB} =$ *kept*$\xrightarrow{\text{oprd}} \mathtt{JJR} =$ *higher*).
c. a superlative adjective ($\mathtt{TAG} = \mathtt{JJS}$);
Example: ...central banks in the west are keeping asset prices the lowest...
($\mathtt{ToI} =$ *central banks*$\rightarrow \mathtt{VERB} =$ *keeping*$\xrightarrow{\text{oprd}} \mathtt{JJS} =$ *lowest*).

ToI connected to a verb followed by an adverbial modifier

The ToI is associated with a verb ($\mathtt{POS} = \mathtt{VERB}$) that is followed by an adverbial modifier relation ($\mathtt{DEP} = \mathtt{advmod}$); this means that it connects the verb to a non-clausal adverb or adverbial phrase that serves to modify the predicate. The adverb ($\mathtt{POS} = \mathtt{ADV}$) can be in the form of:

a. a comparative adverb ($\mathtt{TAG} = \mathtt{RBR}$);
Example: ...expectations for Q1 are that demand will increase further...
($\mathtt{ToI} =$ *demand*$\rightarrow \mathtt{VERB} =$ *increase*$\xrightarrow{\text{advmod}} \mathtt{RBR} =$ *further*).
b. a superlative adverb ($\mathtt{TAG} = \mathtt{RBS}$);
Example: ...monetary policy is best focused on the stability of the general level of prices... ($\mathtt{ToI} =$ *monetary policy*$\rightarrow \mathtt{VERB} =$ *focused*$\xrightarrow{\text{advmod}} \mathtt{RBS} =$ *best*).

ToI connected to a verb associated with a noun

In this case, the ToI is connected to a verb ($\mathtt{POS} = \mathtt{VERB}$) that is linked to a noun ($\mathtt{POS} = \mathtt{NOUN}$) by means of one of the following relations:

a. a direct object ($\mathtt{DEP} = \mathtt{dobj}$), i.e., a clause that connects a transitive verb to a nominal representing the recipient of the action of such predicate;
Example: ...in this quarter profits reached the bottom... ($\mathtt{ToI} =$ *profits*$\rightarrow \mathtt{VERB} =$ *reached*$\xrightarrow{\text{dobj}} \mathtt{NOUN} =$ *bottom*).
b. an attribute ($\mathtt{DEP} = \mathtt{attr}$), i.e., a clause that connects a copula verb to a noun being the non-verb phrase predicate of such verb;
Example: ...in France gold has long been a secure investment...
($\mathtt{ToI} =$ *gold*$\rightarrow \mathtt{VERB} =$ *been*$\xrightarrow{\text{attr}} \mathtt{NOUN} =$ *investment*).

ToI connected to a verb followed by a prepositional modifier

The ToI is connected to a verb ($\mathtt{POS} = \mathtt{VERB}$) that is followed by a prepositional phrase that modifies the heading verb ($\mathtt{DEP} = \mathtt{prep}$). The prepositional modifier is linked to an adposition ($\mathtt{POS} = \mathtt{ADP}$) that establishes a grammatical relationship between the complement and another word or phrase in the context. An adposition typically creates a semantic relationship which may be spatial (in, on, under, etc.), temporal (after, during, etc.), or of some other type (of, for, via, etc.). The adposition is then connected to one of the following terms:

a. a noun ($\mathtt{POS} = \mathtt{NOUN}$), by means of an object of preposition relation ($\mathtt{DEP} = \mathtt{pobj}$), i.e., a noun phrase that follows a preposition and completes its meaning;
Example: ...the German economy is in great shape... ($\mathtt{ToI} =$ *economy*$\rightarrow \mathtt{VERB} =$ *is*$\xrightarrow{\text{prep}} \mathtt{ADP} =$ *in*$\xrightarrow{\text{pobj}} \mathtt{NOUN} =$ *shape*).
b. a verb ($\mathtt{POS} = \mathtt{VERB}$), by means of a complement of preposition relation ($\mathtt{DEP} = \mathtt{pcomp}$), i.e., a clause that is not a pobj and directly connects the preposition with any dependent completing its meaning;
Example: ...the company is increasing its business after expanding its line of products... ($\mathtt{ToI} =$ *company*$\rightarrow \mathtt{VERB} =$ *increasing*$\xrightarrow{\text{prep}} \mathtt{ADP} =$ *after*$\xrightarrow{\text{pcomp}}$
$\mathtt{NOUN} =$ *expanding*).

3.2. Sentiment polarity scoring and propagation

Semantic rules are the instrument that we use to isolate the sentiment connotation of a term of interest. These rules can be considered as building blocks in the sense that they can be concatenated to enrich linguistic analysis. Modelling the complexity of the human language is a challenging task: our goal with these rules is to capture linguistic regularities that can provide a reasonably accurate representation of sentiment regarding a specific topic of interest.

The outcome of the rules consists of a set of words that are semantically related to the term of interest and that we assume are relevant to consider in determining the sentiment of the sentence. We assign a sentiment polarity score to these words – based on the SentiBigNomics dictionary – which is then propagated to the root search concept from the last actor involved in the chunk. Assume that we have two consecutive terms, namely $\mathtt{y}$ and $\mathtt{x}$, with their sentiment scores denoted by $s(\mathtt{y})$ and $s(\mathtt{x})$, respectively. We propagate the sentiment score from the last actor identified by the rule, say $\mathtt{x}$, to its previous linked term, $\mathtt{y}$, to obtain the propagated sentiment score $\widehat{s}(\mathtt{y})$ defined as follows:

$$s(\mathtt{x}) \rightarrow s(\mathtt{y}) \Longrightarrow \widehat{s}(\mathtt{y}). \tag{4}$$

More specifically, we assume that the magnitude of the propagated score, $\widehat{s}(\mathtt{y})$, is given by:

$$\widehat{s}(\mathtt{y}) = sign(s(\mathtt{y})) * sign(s(\mathtt{x})) * (|s(\mathtt{y})| + (1 - |s(\mathtt{y})|) * |s(\mathtt{x})|). \tag{5}$$

The rationale behind this formula is to modify the sentiment score for y proportionally to the sentiment score of the subsequent linked term, x.

For example, the text fragment "*a great value*" is consistent with the rule *value* (0.2) → *great* (0.7), with the numbers in brackets representing the sentiment scores. The score for *great* propagates to *value*, that is 0.7 → 0.2, and, according to Eq. (5), the new sentiment for *value* is given by $(0.2 + (1 - 0.2) * 0.7) = 0.76$. Since the two terms have the same sign, the sentiment score for *value* is increased proportionally with respect to the sentiment of *great*.

The opposite situation occurs when the linked terms have scores with opposite signs. One example is given by the text fragment "*a bad deal*" with the rule linking *deal*(0.2) → *bad*(−0.5). The sentiment score for *bad* propagates to *deal*, that is −0.5 → 0.2, and, according to Eq. (5), the resulting sentiment score for *deal* is negative and equal to $-(0.2 + (1 - 0.2) * 0.5) = -0.6$.

Another situation is when we have two terms that have negative sentiment scores, for example in the text fragment "*unemployment is decreasing*". In this case the linguistic rules detect the link *unemployment*(−0.7) → *decreasing*(−0.4). The sentiment score for *decreasing* propagates to *unemployment*, that is −0.4 → −0.7, which results in a positive sentiment score for *unemployment* and equal to $(0.7 + (1 - 0.7) * 0.4) = 0.82$.

We believe the adopted scoring formula represents an appropriate general rule of thumb for calculating and propagating the sentiment polarities within the selected chunks. In case of concatenation of more rules within the same text fragment, the overall chunk sentiment is computed by using the methodology presented above. In particular, we associate the corresponding sentiment polarity score to each term with the activated rules and calculate the sentiment score for each rule. Finally, we propagate the aspect-based polarity score to the triggering ToI. Note also that the sentiment score for the ToI does not enter into the overall computation, but only its sign is taken into consideration. In particular, if the ToI has a negative connotation, then the final chunk sentiment is inverted.

Suppose, for instance, that the following generic rule is active: ToI→a→b→c. As discussed earlier, the sentiment score propagates from the right-most term, c, to the triggering ToI, as follows:

$$s(\mathtt{c}) \rightarrow s(\mathtt{b}) \rightarrow s(\mathtt{a}) \rightarrow s(\mathtt{ToI}). \tag{6}$$

Following this chain, the last actor, c, propagates its sentiment score $s(\mathtt{c})$ to the previous linked term, b, with sentiment $s(\mathtt{b})$, resulting in a new sentiment polarity, $\widehat{s}(\mathtt{b})$. In this way, Eq. (6) reduces to $\widehat{s}(\mathtt{b}) \rightarrow s(\mathtt{a}) \rightarrow s(\mathtt{ToI})$. The process is iterated with $\widehat{s}(\mathtt{b})$ propagating to $s(\mathtt{a})$ and producing a new polarity score $\widehat{s}(\mathtt{a})$ that reduces Eq. (6) to $\widehat{s}(\mathtt{a}) \rightarrow s(\mathtt{ToI})$. The final iteration provides a propagated score for the ToI denoted by $\widehat{s}(\mathtt{ToI})$. In summary, sentiment propagation across rules works as follows:

$$s(\mathtt{c}) \rightarrow s(\mathtt{b}) \rightarrow s(\mathtt{a}) \rightarrow s(\mathtt{ToI}) \Longrightarrow$$

$$\Longrightarrow \widehat{s}(\mathtt{b}) \rightarrow s(\mathtt{a}) \rightarrow s(\mathtt{ToI}) \Longrightarrow$$

$$\Longrightarrow \widehat{s}(\mathtt{a}) \rightarrow s(\mathtt{ToI}) \Longrightarrow$$

$$\Longrightarrow \widehat{s}(\mathtt{ToI}).$$

We now provide an example of the application of the algorithm on an entire sentence. The inputs of the algorithm are a ToI (e.g., "industrial production") and a location (e.g., "United States"). After parsing the input text, suppose that the algorithm detects the following sentence: "*US manufacturing sector is suffering the biggest decline since World War II*" (see Fig. 3 for an illustration of the text fragment).

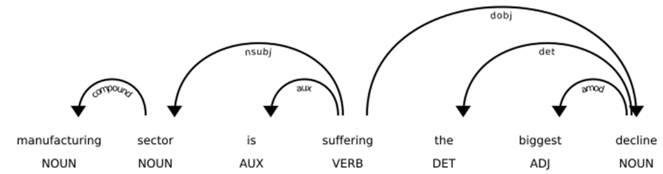


Fig. 3. An illustration of the focused fragment of text for the sentence: *US manufacturing sector is suffering the biggest decline since World War II.*

The algorithm recognises that the sentence is related to the "United States". In addition, by querying the WBG Ontology, the algorithm recognises that "manufacturing sector" is a synonym for "industrial production". Therefore, the sentence is selected as a candidate for the FiGAS sentiment extraction, and the following steps are performed:

1. **Input** ⇒ *US manufacturing sector is suffering the biggest decline since World War II*;
2. The ToI *manufacturing sector* is detected by looping on the part-of-speech tags from the sentence (see Fig. 3);
3. The ToI is linked to the verb *suffering* (POS = VERB), which is related to the noun *decline* (POS = NOUN);
4. VERB and NOUN are linked by means of a DEP = dobj (i.e., direct object) relation;
5. The rule *"ToI connected to a verb associated with a noun"* is triggered: ToI = *manufacturing sector* → VERB = *suffering* $\xrightarrow{\text{dobj}}$ NOUN = *decline*;
6. The NOUN *decline* is linked to the superlative adjective *biggest* (POS = ADJ and TAG = JJS);
7. NOUN and ADJ are linked by a DEP = amod (i.e., adjectival modifier) relation;
8. The NOUN *decline* further triggers the rule *"ToI associated with an adjectival modifier"* ToI = *decline* $\xrightarrow{\text{amod}}$ JJS = *biggest*;
9. The two activated rules concatenate as follows (*rule concatenation*):

   ToI = *manufacturing sector* → VERB = *suffering* $\xrightarrow{\text{dobj}}$ NOUN = *decline* $\xrightarrow{\text{amod}}$ JJS = *biggest*;
10. The sentiment scores for the terms, taken from the SentiBigNomics dictionary, are propagated to the root ToI (*manufacturing sector*) from the last actor involved in the chunk (*biggest*), as described in Eq. (6) and iteratively using Eq. (5) (*polarity propagation*). More specifically:

    i. $s(biggest) = 0.125 \rightarrow s(decline) = -0.6 \rightarrow s(suffering) = -0.5 \rightarrow s(manufacturing\ sector) = 0 \Longrightarrow$
    ii. $\Longrightarrow \widehat{s}(decline) = -0.65 \rightarrow s(suffering) = -0.5 \rightarrow s(manufacturing\ sector) = 0 \Longrightarrow$
    iii. $\Longrightarrow \widehat{s}(suffering) = -0.825 \rightarrow s(manufacturing\ sector) = 0 \Longrightarrow$
    iv. $\Longrightarrow \widehat{s}(manufacturing\ sector) = -0.825$.
11. **Output** ⇒ The final aspect-based sentiment polarity of the ToI *manufacturing sector* in the context of the considered chunk is thus **−0.825**.

4. Computational analysis

4.1. Algorithms comparison

In this section, we compare the performance and efficiency of the following lexicon-based SA algorithms that were discussed earlier in the paper:

- *FiGAS*, for the proposed Fine-Grained Aspect-based Sentiment analysis approach (Section 3);
- *SentiWNet*, for the coarse-grained SA method using the SentiWordNet lexicon (Section 2.1);
- *Sentic*, for the coarse-grained SA method using the SenticNet lexicon (Section 2.1);
- *Harvard*, for the coarse-grained SA method using the Harvard IV-4 lexicon (Section 2.3);
- *LMD*, for the coarse-grained SA method using the Loughran and McDonald lexicon (Section 2.4);
- *SentiBigNomics*, for the coarse-grained SA method using our SentiBigNomics lexicon (Section 2.5).

In our experiments we compare the sentiment scores obtained by the different SA algorithms with respect to the scores provided by human annotation. In particular, we select a set of English sentences in the economic and financial domains from the commercial Dow Jones Data, News and Analytics (DNA) platform.[20] We consider only sentences in the English language that were published in a set of popular US and UK newspapers[21] from January 1980 until the end of 2020. In our dataset we only exclude articles that are classified by Dow Jones DNA in the sport news category.

We select sentences in the economic and financial contexts by performing a keyword search with respect to a list of 29 `ToI` regarding various aspects of economic activity.[22] In addition, we consider bigrams for each `ToI` in order to increase the probability that the sentences specifically relate to a facet of the state of the economy. The resulting dataset is composed of 102,471 sentences. After computing the *FiGAS* sentiment scores, we randomly sample 1400 sentences to be representative of the complete range of sentiment values (i.e., $[-1, +1]$). We do this by constraining the sentiment scores to follow a uniform distribution $U(-1, +1)$.[23] Based on this sample of 1400 sentences, we calculate the sentiment scores of the various lexicon-based SA methods discussed above.

The benchmark for the considered SA methods is the human annotation of the sentences performed by nine US-based professional annotators via the AWS SageMaker service. We collect the sentiment polarity scores expressed by the annotators for each sentence and the indicated economic topic of interests. In particular, given each sentence and the corresponding `ToI`, the labelling instructions explicitly ask the annotators to "select the sentiment score that the sentence conveyed with respect to the indicated topic". The score magnitude represents the sentiment intensity of the sentence with respect to the `ToI` as perceived by the human annotator; for instance, a score close to 1 indicates that the sentence expresses a very good economic situation with respect to the topic of interest, while a positive value close to 0 represents a moderately favourable situation. The annotators are asked to assign the sentiment score on a fixed scale of values between -1 and 1 at intervals of 0.2 (i.e., the possible values are $\{-1, -0.8, -0.6, -0.4, -0.2, 0, 0.2, 0.4, 0.6, 0.8, 1\}$). In case they believe that a sentence is neutral regarding the economic `ToI`, they have the option to assign the null score to the sentence.

We calculate the median and average scores assigned by the annotators for each sentence, as well as the standard deviation as proxy for the inter-annotator agreement. As expected, we observe large variability in scores across annotators, as confirmed by the 0.5 average standard deviation of the annotator scores. Following [60], we calculate the κ measure of agreement among the scores assigned by human annotators, in particular using the variant by Fleiss et al. [61] to account for multiple raters, where κ is defined as:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{7}$$

where p_o is the observed agreement between annotators, while p_e is the agreement level that we would expect if the annotators are choosing the answer randomly. The κ measure is defined at $[-1, +1]$, with negative values indicating poor agreement and positive ones signalling agreement. Considering a uniform distribution among the categories, we obtain a $\kappa = 0.03$ (p-value $= 0$), which indicates only slight agreement between annotators. While small, the κ value is significantly different to zero (i.e., the agreement is statistically different from the one that we could have obtained randomly) since the p-value is extremely small. Obtaining a clear agreement on a set of 11 possible scores is certainly a difficult task, as the annotation job and the choice of the score magnitude are subjective in nature. However, we expect that annotators would, to a larger degree, agree on the sign of the sentiment scores. For this reason, we also calculate the κ measure considering only 3 possible labels corresponding to the sentiment score indications $\{-1, 0, +1\}$. Considering a uniform distribution among the categories, we obtain a $\kappa = 0.16$ (p-value $= 0$), which indicates higher agreement between annotators on whether the sentiment expressed by the sentence is positive, negative or neutral for the `ToI`. Moreover, the associated p-value is zero, so we can reject the null hypothesis that this agreement level is the same as we would have obtained by chance. If we consider the positive and negative classes only, the κ score further increases to 0.43 (with a p-value $= 0$), indicating an overall moderate agreement among the annotators, and a higher confidence in the choice of these classes relative to the neutral sentiment category, which appears to yield more uncertainty among the raters.

We now compare the median score of the nine human annotators to the sentiment score assigned by the considered SA algorithms at sentence level. The advantage of considering the median score is that it is robust to extreme score choices from some annotators, in addition to the fact that it provides a value belonging to the score categories. For completeness, we also report in the Appendix the same analysis performed by using the average and modal scores as the ground truth sentiment polarity, which appear to be in line with the results obtained with the median scores, detailed in the following.

The first measure that we consider in the comparison is the count of the number of cases in which the sentiment score given by an algorithm is the closest to the human annotation. The resulting ranking is as follows (from the best to the worst): *FiGAS* $=$ 495, *LMD* $=$ 277, *SentiBigNomics* $=$ 224, *SentiWNet* $=$ 224, *Sentic* $=$ 223, *Harvard* $=$ 207. The results show that the increased accuracy of *FiGAS* is remarkable and provides the closest predicted scores to the human annotated ones. The other algorithms perform similarly, with a slightly better performance by *LMD* compared to the remaining algorithms. If we calculate the average absolute difference between the algorithmic and human annotated scores, we obtain the following results (from the best to the worst): *FiGAS* $= 0.403$, *SentiBigNomics* $= 0.410$, *SentiWNet*

[20] Dow Jones DNA, https://professional.dowjones.com/developer-platform/.

[21] US newspapers: The New York Times, Wall Street Journal, Washington Post, Dallas Morning News, San Francisco Chronicle, and Chicago Sun-Times. UK newspapers: The Times, The Guardian, Daily Mail, The Economist, Evening Standard, The Sunday Times and Observer.

[22] The `ToI` considered are: "industry index", "capital growth", "capital return", "capital activity", "capital industry", "industrial index", "automobile profit", "industry profit", "capital profit", "capital output", "construction index", "production return", "turnover profit", "capital index", "capital production", "capital sector", "construction profit", "industry return", "agricultural index", "productive return", "factory profit", "construction return", "factory return", "agricultural profit", "automobile index", "manufacturing profit", "productive profit", "agricultural return", "secondary return".

[23] In order to sample under the $U(-1, +1)$ distribution, we divided the scores in ten bins of size 0.2, and then randomly selected 140 sentences in each bin.

Table 3
Pairwise differences (absolute values) of the average algorithm ranks by using the median score of the nine annotators*.

Method (rank)	*FiGAS* (3.256)	*SentiBigNomics* (3.463)	*LMD* (3.483)	*SentiWNet* (3.484)	*Sentic* (3.564)	*Harvard* (3.757)
FiGAS (3.256)	–	**0.207**	**0.227**	**0.228**	**0.308**	**0.501**
SentiBigNomics (3.463)	**0.207**	–	0.020	0.021	0.101	**0.294**
LMD (3.483)	**0.227**	0.020	–	0.001	0.081	**0.274**
SentiWNet (3.484)	**0.228**	0.021	0.001	–	0.080	**0.273**
Sentic (3.564)	**0.308**	0.101	0.081	0.080	–	**0.193**
Harvard (3.757)	**0.501**	**0.294**	**0.274**	**0.273**	**0.193**	–

* Critical difference = 0.183 for a significance level of 1% for the Nemenyi test.

= 0.424, *Sentic* = 0.425, *LMD* = 0.431, *Harvard* = 0.455. Again, the results clearly confirm the superiority of the *FiGAS* approach with respect to the other algorithms since it delivers the smallest absolute deviation from the human annotation. While the deviations of the remaining algorithms are very close to one another, we observe that *Harvard* performs worst in all measures.

We now report some illustrative examples consisting of sentences from the dataset that provide a qualitative view of the characteristics of the different algorithms. For instance, consider the `ToI` *"industrial production"*, and the sentence *On Tuesday, investors learned that German industrial production fell 4% in August since July, the largest such decline since 2009*. This sentence has an obvious negative connotation towards the `ToI`, as also expressed by the human annotators that provide a median polarity score equal to −0.6. *FiGAS* assigns a −0.6 score – similar to the human annotators – whereas the score for the other algorithms range between 0 (for *LMD* and *SentiBigNomics*) and 0.1 (all others). The positive score assigned by some algorithms is due to the misleading effect of the term *"largest"* that has a large and positive tone. However, *FiGAS* correctly filters such information and correctly propagates an appropriate sentiment polarity to the `ToI`, as assessed by the human annotators.

Consider the following sentence as another example: *Overall, manufacturing rose by a seasonally adjusted 0.6 percent, although that was driven largely by a surge in utility prices during a stretch of cold weather in December.* For the `ToI` *"manufacturing"*, this sentence seems to provide positive sentiment regarding the `ToI`. This is confirmed by the human annotators that express a median polarity score equal to 0.4. *FiGAS* assigns a score of 0.5, while the score from the other algorithms range between −0.1 (*Harvard*) and 0.2 (*SentiBigNomics* and *SentiWNet*). Also in this case, the *FiGAS* algorithm demonstrates to be the most consistent with the human annotated scores. *SentiBigNomics* and *SentiWNet* also behave fairly well – providing a positive score – although not as close as that of *FiGAS*. *LMD* and *Sentic* assign a null score, probably misled by the terms *"stretch"* and *"cold"*. This error is even more evident with *Harvard*, which assigns a slightly negative sentiment score of −0.1, and provides the worst performance in this case also. Both of these illustrative examples show the higher accuracy of a fine-grained, aspect-based SA approach compared to alternative SA methods. The results also confirm the ability of the FiGAS algorithm to measure the natural understanding of the text and to better reflect human reasoning.

In order to provide a more robust comparison of the performance of the different algorithms, we extend the earlier statistical analysis of the results. First, we rank each SA method based on the closeness of the algorithmic score to the human score. We then average these ranks and the ordering is as follows (from the best to the worst rank): *FiGAS* = 3.256, *SentiBigNomics* = 3.463, *LMD* = 3.483, *SentiWNet* = 3.484, *Sentic* = 3.564, *Harvard* = 3.757.

We use the *Friedman Test* [62,63] and its corresponding *Nemenyi Post-hoc Test* [64,65] to evaluate the statistical significance of differences in ranks. Madjarov et al. [66] and Demśar [67] provide details on statistical tests used for algorithms comparison with multiple datasets. Based on the Friedman Test, we find a significant difference between the evaluated ranks (at the 1% significance level). Since the null hypothesis of equivalence in algorithm rankings is rejected, we also perform a pairwise comparison using the Nemenyi post-hoc test. This test considers the performance of two algorithms significantly different if the difference in average ranks is greater than a threshold critical difference, which – at 1% level – corresponds to the critical value of 0.183.

The differences between the average ranks of the algorithms are reported in Table 3. The results suggest that *FiGAS* is statistically different from the rest of the methods according to the Nemenyi test, since the pairwise differences of its average rank with respect to the other ranks (in bold in Table 3) are all larger compared to the critical value of 0.183. When comparing the ranks of *SentiBigNomics*, *LMD*, *SentiWNet* and *Sentic*, all of the differences between them are smaller than the critical difference, 0.183, meaning that their performance is statistically comparable. However, the differences in their average ranks with respect to that of *Harvard*, highlighted in bold in Table 3, are greater compared to the critical value, which suggests that they statistically outperform *Harvard* based on the Nemenyi test. The mediocre performance of *Harvard* is probably due to the fact that the Harvard IV-4 dictionary was originally developed for psychological applications and may not be well-suited to analyse text in the economic and financial domains.

To summarise, the statistical evaluation of the algorithms shows that the novel *FiGAS* approach consistently outperforms the other lexicon-based SA approaches. According to our computational experience, the proposed method shows how to better adapt to the specific economic and financial domains, assigning a similar sentiment score as human annotators in the vast majority of cases, and confirming the superiority of a fine-grained SA strategy with respect to coarse-grained SA methods.

4.2. Application: economic sentiment indicators

There are several studies that extract sentiment from social media, newspapers and microblogs with the goal of forecasting and nowcasting economic and financial variables (see [1,12–14], among others). In this section we provide an application related to the deployment of *FiGAS* for the construction of economic sentiment indicators of the current state of the economy based on two large textual datasets, namely the US Federal Reserve (FED) Beige Book[24] and the European Central Bank (ECB) Monthly Bulletin.[25] In order to provide an illustrative comparison of the performance of *FiGAS* with respect to a coarse-grained SA approach, we also calculate an indicator based on the *SentiBigNomics* lexicon. We choose to use *SentiBigNomics* in order to explore the

[24] FED Beige Book, available at: https://www.federalreserve.gov/monetarypolicy/beige-book-default.htm.

[25] ECB Bulletin, available at: https://www.ecb.europa.eu/pub/economic-bulletin/mb/html/index.en.html.

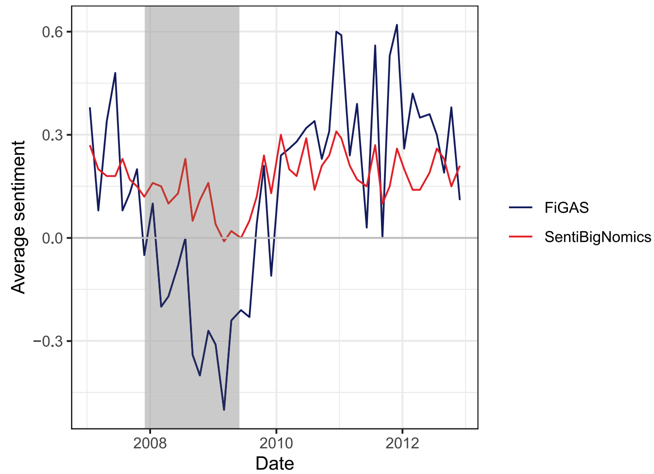


Fig. 4. FED Beige Book sentiment indicator from 2007 to 2013 computed using the FiGAS algorithm (blue) and the SentiBigNomics approach (red). The shadowed grey area corresponds to the recessionary period following the NBER business cycles reference dates.

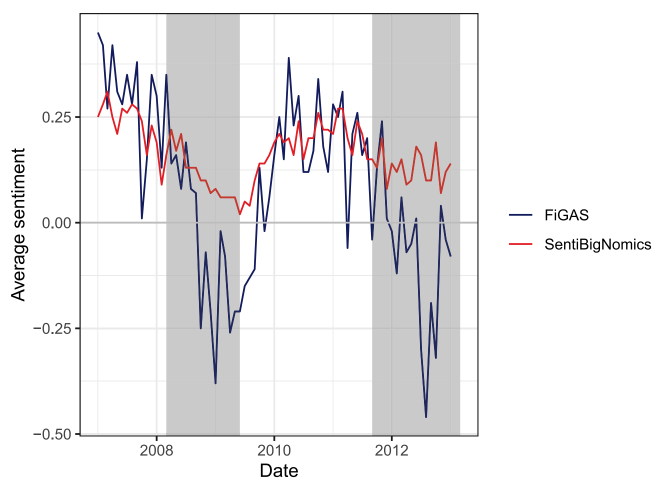


Fig. 5. ECB Bulletin sentiment indicator from 2007 to 2013 computed using *FiGAS* (blue) and *SentiBigNomics* (red). The shadowed grey area corresponds to the recessionary period following the EABCN business cycles reference dates.

added value of the linguistic rules developed in *FiGAS* while controlling for the underlying lexical resource. As shown in Table 3, *SentiBigNomics* has performed the best among the coarse-grained methods explored.

With regard to the datasets, the FED Beige Book is released eight times per year, while the ECB Bulletin is available at monthly intervals. Both sources provide general information about the current state of the economy in the respective territories. In particular, the FED Beige Book discusses the current state of the national and regional economies based on the information gathered by the regional Federal Reserve Banks. The ECB Bulletin instead reports a comprehensive analysis of the economic and monetary developments in the European Union, which serves as the basis for policy-makers' decisions. Both datasets are publicly available and their relevant potential for economic predictions has already been explored in the literature [68,69]. To provide an illustrative application of the proposed algorithm, we consider reports released between January 2007 and January 2013. We use the FED Beige Book and the ECB Bulletin as input for our *FiGAS* algorithm and build a sentiment indicator for the `ToI` *"economic activity"*.

Fig. 4 plots the sentiment indicators computed on the releases of the FED Beige Book using the two alternative methods: in blue, the sentiment indicator obtained using the *FiGAS* algorithm, while in red the *SentiBigNomics* sentiment indicator, calculated by averaging the sentiment scores assigned by the SentiBigNomics lexicon (Section 2) to each word in the document. The grey shadowed area corresponds to the recessionary period as identified by the National Bureau of Economic Research (NBER) business cycles reference dates.[26] The *FiGAS* sentiment indicator seems to correctly capture the recessionary period; as of 2008, we observe a steep drop in the *FiGAS* sentiment indicator, which takes negative values, hence indicating an unfavourable economic situation. On the other hand, the *SentiBigNomics* sentiment indicator does not properly capture the recessionary period. While it shows a decreasing trend in the first part of the sample, the *SentiBigNomics* sentiment indicator only reaches negative values in the first months of 2009.

Similarly, in Fig. 5 we report the *FiGAS* and *SentiBigNomics* sentiment measures built on the ECB Bulletin. The shadowed areas indicate the double-dip recessions that hit Europe in 2008 and 2011, as detailed by the Euro Area Business Cycle Network (EABCN) business cycles reference dates.[27] The *FiGAS* sentiment indicator promptly reacts to both recessions, indicating the deterioration of economic activity in Europe. On the other hand, the *SentiBigNomics* sentiment indicator barely captures the 2008 recession and – even to a minor extent – the 2011 sovereign debt crisis. Overall, the *FiGAS* sentiment indicator seems to more accurately, and in a timelier manner, capture the actual state of the economy by leveraging the set of semantic rules to extract a fine-grained, aspect-based sentiment score for the specific topic of interest. This qualitative analysis further shows the superiority of the proposed methodology. In addition, *FiGAS* offers the option to detect the tense of the constructed term chunks and to filter out only the sentiment scores of one or more indicated verb tenses, in order to explore the effects of the sentiment expressed for an economic concept in a given text for specific verbal forms. This, among other potential applications of *FiGAS* for economic forecasting and nowcasting, is currently under investigation in our ongoing works related to the prediction of macroeconomic variables for the US and the major European economies.

5. Conclusions and future work

The extraction of sentiment from text in financial and economic applications has been the subject of great attention in recent years. In this paper, we focus on lexicon-based SA methods that consist of unsupervised approaches relying on an external dictionary of sentiment words. In addition, we propose *FiGAS* – a novel Fine-Grained Aspect-based Sentiment analysis method – as an alternative approach to SA. The proposed algorithm relies on a set of semantic rules that are able to identify the sentiment associated with specific topics of interest within the text of each sentence in a document. The algorithm is *fine-grained* as it assigns a real-valued sentiment polarity score in the range $[-1, +1]$, and is *aspect-based* as the sentiment specifically refers to the token of interest. Polarity scores for individual words are taken from a humanly annotated lexicon, referred to as SentiBigNomics, that has been designed for applications in the financial and economic domains.

In order to evaluate the performance of the proposed algorithm, we design a humanly annotated dataset of sentences

26 NBER business cycles reference dates can be consulted at https://www.nber.org/cycles.html.

27 EABCN business cycles reference dates can be consulted at https://eabcn.org/dc/chronology-euro-area-business-cycles.

referring to various aspects of the state of the economy which we use for a statistical comparison of the performance of FiGAS with other popular lexicon-based SA approaches. Our FiGAS algorithm shows very promising results, by consistently outperforming the other methods with statistical evidence, and assigning a sentiment score closer to the human score in the largest number of cases. Overall, FiGAS seems to accurately capture the sentence mood for the specific economic topic of interest, and the resulting sentiment indicators evaluated based on the FED Beige Book and the ECB Bulletin data show the potential for the use of FiGAS in economic and financial forecasting applications. In an ongoing work, we aim to assess the validity of the proposed algorithm in building sentiment indicators for various European countries and the US, testing the usefulness of the computed sentiment indicators for forecasting real-world economic and financial variables.

CRediT authorship contribution statement

Sergio Consoli: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Investigation, Writing – review & editing. **Luca Barbaglia:** Software, Data curation, Formal analysis, Writing – original draft, Visualization, Investigation, Writing – review & editing. **Sebastiano Manzan:** Visualization, Investigation, Formal analysis, Writing – original draft, Validation, Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The author Dr. Sergio Consoli wishes to dedicate this work with deepest respect to the memory of Professor Kenneth Darby-Dowman, a great scientist, an excellent manager, the best supervisor, a wonderful person, a real friend.

Appendix

We evaluate the robustness of our findings we also evaluate the experimental results obtained using the average and modal scores of the nine human annotators as the ground truth sentiment polarity. As shown below, the results appear to be very similar to those obtained using the median scores, detailed in Section 4.1.

A. Average score of the nine human annotators

We first look at the average score of the nine human annotators, and compare it to the sentiment score assigned by the considered SA algorithms at the sentence level. If we count the number of cases in which the sentiment score given by an algorithm is the closest to the human annotation, we obtain the following ranking of the methods(from the best to the worst): *FiGAS* = 487, *LMD* = 274, *Sentic* = 220, *Harvard* = 215, *SentiBigNomics* = 201, *SentiWNet* = 199. Similarly to the median, also the results obtained by considering the average score of the nine human annotators show a remarkable performance of *FiGAS* relative to the other algorithms, providing the closest predicted scores to the human annotated ones. The other algorithms perform similarly, with a slightly better performance of *LMD* compared to the remaining algorithms, as happened also by considering the median score. If we calculate the average absolute difference between the algorithmic and human annotated scores, we obtain the following results (from the best to the worst): *FiGAS* = 0.388, *SentiBigNomics* = 0.408, *SentiWNet* = 0.425, *Sentic* = 0.425, *LMD* = 0.430, *Harvard* = 0.455. Similarly to what happened when considering the median score as the ground truth sentiment polarity, also the results obtained by using the average score of the nine human annotators clearly confirm the superiority of the *FiGAS* approach with respect to the other algorithms, since it delivers the smallest absolute deviation from the human annotation. While the deviations of the remaining algorithms are very close to one another, we observe that *Harvard* performs worst in all measures, as detected also by using the median score.

As in Section 4.1, we also perform the statistical tests of Friedman and Nemenyi to evaluate the statistical significance of differences in ranks among the algorithms. We use the average score of the nine human annotators as the ground truth sentiment polarity. First, by ranking each SA method based on the closeness of the algorithmic score to the human score, we obtain the following average ranks (from the best to the worst rank): *FiGAS* = 3.160, *SentiWNet* = 3.473, *LMD* = 3.480, *SentiBigNomics* = 3.507, *Sentic* = 3.591, *Harvard* = 3.788.

By using the Friedman Test, we find a significant difference between the evaluated ranks, at the 1% significance level. Since the null hypothesis of equivalence in algorithm rankings is rejected, we perform a pairwise comparison using the Nemenyi post-hoc test, whose threshold critical difference at 1% level corresponds to the critical value of 0.183. The differences between the average ranks of the algorithms are reported in Table 4. In line with the results in Section 4.1, we find *FiGAS* to be statistically superior with respect to all the other methods according to the Nemenyi test, since the pairwise differences of its average rank with respect to the other ranks (in bold in Table 4) are all larger than the critical value of 0.183. We also find there is a group of statistically comparable algorithms, i.e. *SentiWNet*, *LMD*, *SentiBigNomics*, and *Sentic*, whose pairwise differences result to be smaller than the critical difference, 0.183. Last, *Harvard* obtains the worst performance, which is statistically different with respect to all the other algorithms according to the Nemenyi test.

Table 4
Pairwise differences (absolute values) of the average algorithm ranks by using the average score of the nine annotators*.

Method (rank)	*FiGAS* (3.160)	*SentiWNet* (3.473)	*LMD* (3.480)	*SentiBigNomics* (3.507)	*Sentic* (3.591)	*Harvard* (3.788)
FiGAS (3.160)	–	**0.313**	**0.320**	**0.347**	**0.431**	**0.628**
SentiWNet (3.473)	**0.313**	–	0.007	0.034	0.118	**0.315**
LMD (3.480)	**0.320**	0.007	–	0.027	0.111	**0.308**
SentiBigNomics (3.507)	**0.347**	0.034	0.027	–	0.084	**0.281**
Sentic (3.591)	**0.431**	0.118	0.111	0.084	–	**0.197**
Harvard (3.788)	**0.628**	**0.315**	**0.308**	**0.281**	**0.197**	–

* Critical difference = 0.183 for a significance level of 1% for the Nemenyi test.

Table 5
Pairwise differences (absolute values) of the average algorithm ranks by using the modal score of the nine annotators*.

Method (rank)	*FiGAS* (2.850)	*SentiWNet* (3.537)	*LMD* (3.546)	*SentiBigNomics* (3.573)	*Sentic* (3.645)	*Harvard* (3.849)
FiGAS (2.850)	–	**0.687**	**0.696**	**0.723**	**0.795**	**0.999**
SentiWNet (3.537)	**0.687**	–	0.009	0.036	0.108	**0.312**
LMD (3.546)	**0.696**	0.009	–	0.027	0.099	**0.303**
SentiBigNomics (3.573)	**0.723**	0.036	0.027	–	0.072	**0.281**
Sentic (3.645)	**0.795**	0.108	0.099	0.072	–	**0.204**
Harvard (3.849)	**0.999**	**0.312**	**0.303**	**0.281**	**0.204**	–

* Critical difference = 0.183 for a significance level of 1% for the Nemenyi test.

B. Modal score of the nine human annotators

We then consider the mode of the scores from the nine human annotators, that is, the value that occurs most frequently among the annotations of each sentence, and compare it to the sentiment score assigned by the SA algorithms at the sentence level. If we count the number of cases in which the sentiment score given by an algorithm is the closest to the human annotation, we obtain the following ranking (from the best to the worst): *FiGAS* = 575, *LMD* = 232, *SentiBigNomics* = 203, *Sentic* = 197, *Harvard* = 190, *SentiWNet* = 186. Similarly to the case of the median and the average scores, the results for the modal score show that *FiGAS* provides by far the closest predicted scores to the human annotated ones relative to the other algorithms. The rest of the algorithms perform quite similarly, with a slightly better performance obtained by *LMD*, as happened also with the median and average scores. If we evaluate the average absolute difference between the algorithmic and human annotated scores, we obtain the following results (from the best to the worst): *FiGAS* = 0.319, *SentiBigNomics* = 0.408, *SentiWNet* = 0.424, *Sentic* = 0.425, *LMD* = 0.431, *Harvard* = 0.454. Hence, also using the modal score of the nine human annotators clearly confirms the superiority of the *FiGAS* approach relative to the other algorithms and the findings for the median and average scores. *FiGAS* is able to get the smallest absolute deviation from the human annotation, with the deviations of the other algorithms appearing to be very close to one another. *Harvard*, again, obtains the worst performance.

As in Section 4.1, we find a significant difference between the ranks of the algorithms using the Friedman test at the 1% significance level (from the best to the worst rank): *FiGAS* = 2.850, *SentiWNet* = 3.537, *LMD* = 3.546, *SentiBigNomics* = 3.573, *Sentic* = 3.645, *Harvard* = 3.849.

By performing the Nemenyi post-hoc test, whose threshold critical difference at 1% level corresponds to the critical value of 0.183, we find three groups of statistically different algorithms. From the differences between the average ranks of the algorithms, reported in Table 5, we find that the first group is composed by *FiGAS*, which is statistically superior with respect to all the other methods, since the pairwise differences of its average rank with respect to the other ranks (in bold in Table 5) are larger than the critical value of 0.183. The second group is composed by *SentiWNet*, *LMD*, *SentiBigNomics*, and *Sentic*, whose pairwise differences result to be smaller than the critical difference, 0.183, and therefore their performance are not statistically different. The last group is made by *Harvard*, which obtains the worst performance and is statistically different with respect to all the other algorithms according to the Nemenyi test. As shown, the results obtained by using the mode of the scores from the nine human annotators as the ground truth sentiment polarity appear to be consistent with the results obtained by using the median and the average scores as the ground truth sentiment polarity.

References

[1] F.Z. Xing, E. Cambria, R.E. Welsch, Natural language based financial forecasting: A survey, Artif. Intell. Rev. 50 (1) (2018) 49–73.
[2] A. Algaba, D. Ardia, K. Bluteau, S. Borms, K. Boudt, Econometrics meets sentiment: An overview of methodology and applications, J. Econ. Surv. 34 (3) (2020) 512–547.
[3] S. Agrawal, P. Azar, A.W. Lo, T. Singh, Momentum, mean-reversion and social media: Evidence from StockTwits and Twitter, J. Portf. Manag. 44 (7) (2018) 85–95.
[4] L.A. Thorsrud, Words are the new numbers: A newsy coincident index of the business cycle, J. Bus. Econom. Statist. 38 (2) (2020) 393–409.
[5] B. Liu, Sentiment Analysis: Mining Opinions, Sentiments, and Emotions, Cambridge University Press, 2015.
[6] E. Cambria, S. Poria, A. Gelbukh, M. Thelwall, Sentiment analysis is a big suitcase, IEEE Intell. Syst. 32 (6, 8267597) (2017) 74–80.
[7] M. Dragoni, I. Donadello, E. Cambria, OntoSenticNet 2: Enhancing reasoning within sentiment analysis, IEEE Intell. Syst. 37 (2) (2022) 1–8, (in press).
[8] L. Einav, J. Levin, Economics in the age of big data, Science 346 (6210) (2014) 715–721.
[9] L. Barbaglia, S. Consoli, S. Manzan, Forecasting GDP in Europe with textual data, 2021, Available at SSRN 3898680.
[10] L. Barbaglia, S. Consoli, S. Manzan, Forecasting with economic news, J. Bus. Econom. Statist. (2022) http://dx.doi.org/10.1080/07350015.2022.2060988.
[11] P.C. Tetlock, Giving content to investor sentiment: The role of media in the stock market, J. Finance 62 (3) (2007) 1139–1168.
[12] M.M. Mostafa, More than words: Social networks' text mining for consumer brand sentiments, Expert Syst. Appl. 40 (10) (2013) 4241–4251.
[13] J. Bollen, H. Mao, X. Zeng, Twitter mood predicts the stock market, J. Comput. Sci. 2 (1) (2011) 1–8.
[14] O. Bustos, A. Pomares-Quimbaya, Stock market movement forecast: A systematic review, Expert Syst. Appl. 156 (2020) 113464.
[15] M. Van De Kauter, D. Breesch, V. Hoste, Fine-grained analysis of explicit and implicit sentiment in financial news articles, Expert Syst. Appl. 42 (11) (2015) 4999–5010.
[16] L. Malandri, F. Xing, C. Orsenigo, C. Vercellis, E. Cambria, Public mood–driven asset allocation: The importance of financial sentiment in portfolio management, Cogn. Comput. 10 (6) (2018) 1167–1176.

[17] F.Z. Xing, E. Cambria, Y. Zhang, Sentiment-aware volatility forecasting, Knowl.-Based Syst. 176 (2019) 68–76.

[18] C.-Y. Chang, Y. Zhang, Z. Teng, Z. Bozanic, B. Ke, Measuring the information content of financial news, in: Proceedings of COLING 2016 - 26th International Conference on Computational Linguistics, 2016, pp. 3216–3225.

[19] M. Doms, N.J. Morin, Consumer Sentiment, the Economy, and the News Media, Finance and Economics Discussion Series 2004–51, Board of Governors of the Federal Reserve System (U.S.), 2004.

[20] S. Hansen, M. McMahon, Shocking language: Understanding the macroeconomic effects of central bank communication, J. Int. Econ. 99 (2016) S114–S133.

[21] M. Gentzkow, B. Kelly, M. Taddy, Text as data, J. Econ. Lit. 57 (3) (2019) 535–574.

[22] M.S. Akhtar, A. Ekbal, E. Cambria, How intense are you? Predicting intensities of emotions and sentiments using stacked ensemble [Application notes], IEEE Comput. Intell. Mag. 15 (1, 8956109) (2020) 64–75.

[23] L.A. Thorsrud, Nowcasting using News Topics. Big Data versus Big Bank, Norges Bank Working Paper 20/2016, 2016.

[24] A. Dridi, M. Atzeni, D. Reforgiato Recupero, FineNews: FIne-grained semantic sentiment analysis on financial microblogs and news, Int. J. Mach. Learn. Cybern. 10 (2018) 2199–2207.

[25] M. Taboada, J. Brooke, M. Tofiloski, K. Voll, M. Stede, Lexicon-based methods for sentiment analysis, Comput. Linguist. 37 (2) (2011) 267–307.

[26] D. Garcia, Sentiment during recessions, J. Finance 68 (3) (2013) 1267–1300.

[27] A.H. Shapiro, M. Sudhof, D.J. Wilson, Measuring news sentiment, J. Econom. 228 (2) (2022) 221–243.

[28] C. Zhou, Q. Tianmei, W. Jixiang, Z. Yu, W. Zhihong, G. Yi, Z. Junfeng, Fine-grained sentiment analysis of foreign exchange news, in: 5th International Conference on Information Management (ICIM), 2019, pp. 279–284.

[29] M. Atzeni, A. Dridi, D. Reforgiato Recupero, Fine-grained sentiment analysis on financial microblogs and news headlines, Commun. Comput. Inf. Sci. 769 (2017) 124–128.

[30] B. Clapham, M. Siering, P. Gomber, Popular news are relevant news! How investor attention affects algorithmic decision-making and decision support in financial markets, Inf. Syst. Front. 23 (2) (2021) 477–494.

[31] N. Jing, Z. Wu, H. Wang, A hybrid model integrating deep learning with investor sentiment analysis for stock price prediction, Expert Syst. Appl. 178 (2021) 115019.

[32] M. Li, W. Li, F. Wang, X. Jia, G. Rui, Applying BERT to analyze investor sentiment in stock market, Neural Comput. Appl. 33 (10) (2021) 4663–4676.

[33] E.K.W. Leow, B.P. Nguyen, M.C.H. Chua, Robo-advisor using genetic algorithm and BERT sentiments from tweets for hybrid portfolio optimisation, Expert Syst. Appl. 179 (2021) 115060.

[34] Z. Jin, Y. Yang, Y. Liu, Stock closing price prediction based on sentiment analysis and LSTM, Neural Comput. Appl. 32 (13) (2020) 9713–9729.

[35] Z. He, J. Zhou, H.-N. Dai, H. Wang, Gold price forecast based on LSTM-CNN model, in: Proceedings of the 17th IEEE International Conference on Pervasive Intelligence and Computing, 8890378, 2019, pp. 1046–1053.

[36] S. Merello, A.P. Ratto, L. Oneto, E. Cambria, Ensemble application of transfer learning and sample weighting for stock market prediction, in: Proceedings of the International Joint Conference on Neural Networks (IJCNN), 8851938, 2019, pp. 1–8.

[37] M.S. Akhtar, A. Kumar, D. Ghosal, A. Ekbal, P. Bhattacharyya, A Multilayer perceptron based ensemble technique for fine-grained financial sentiment analysis, in: Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP), 2017, pp. 540–546.

[38] S. Consoli, L. Barbaglia, S. Manzan, Fine-grained, aspect-based semantic sentiment analysis within the economic and financial domains, in: The 2nd IEEE International Conference on Cognitive Machine Intelligence (CogMI), 9319280, 2020, pp. 52–61.

[39] D. Reforgiato Recupero, V. Presutti, S. Consoli, A. Gangemi, A.G. Nuzzolese, Sentilo: Frame-based sentiment analysis, Cogn. Comput. 7 (2) (2015) 211–225.

[40] T.W. Kim, B.R. Routledge, Informational privacy, a right to explanation, and interpretable AI, in: Proceedings of the 2nd IEEE Symposium on Privacy-Aware Computing (PAC), 2018, pp. 64–74.

[41] L. Gilpin, D. Bau, B. Yuan, A. Bajwa, M. Specter, L. Kagal, Explaining explanations: An overview of interpretability of machine learning, in: Proceedings of the 5th IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2019, pp. 80–89.

[42] E. Cambria, A. Hussain, Sentic Computing: A Common-Sense-Based Framework for Concept-Level Sentiment Analysis, Springer Nature, 2015.

[43] S. Poria, E. Cambria, G. Winterstein, G.-B. Huang, Sentic patterns: Dependency-based rules for concept-level sentiment analysis, Knowl.-Based Syst. 69 (1) (2014) 45–63.

[44] R. Johansson, A. Moschitti, Relational features in fine-grained opinion analysis, Comput. Linguist. 39 (3) (2013) 473–509.

[45] Y. Zhang, R. Jin, Z.-H. Zhou, Understanding bag-of-words model: A statistical framework, Int. J. Mach. Learn. Cybern. 1 (1–4) (2010) 43–52.

[46] T. Loughran, B. McDonald, Measuring readability in financial disclosures, J. Finance 69 (4) (2014) 1643–1671.

[47] S. Baccianella, A. Esuli, F. Sebastiani, SentiWordNet 3.0: An enhanced lexical resource for sentiment analysis and opinion mining, in: LREC, Vol. 10, 2010, pp. 2200–2204.

[48] A. Budanitsky, G. Hirst, Evaluating wordnet-based measures of lexical semantic relatedness, Comput. Linguist. 32 (1) (2006) 13–47.

[49] D. Rao, D. Ravichandran, Semi-supervised polarity lexicon induction, in: Proceedings of the 12th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2009, pp. 675–682.

[50] A. Andreevskaia, S. Bergler, Mining WordNet for fuzzy sentiment: Sentiment tag extraction from WordNet glosses, in: Proceedings of the 11th Conference of the European Chapter of the Association for Computational Linguistics (EACL), 2006, pp. 209–216.

[51] E. Cambria, R. Speer, C. Havasi, A. Hussain, SenticNet: A Publicly available semantic resource for opinion mining, in: 2010 AAAI Fall Symposium Series, Vol. FS-10-02, 2010, pp. 14–18.

[52] Y. Susanto, A.G. Livingstone, B.C. Ng, E. Cambria, The Hourglass model revisited, IEEE Intell. Syst. 35 (5, 9237283) (2020) 96–102.

[53] E. Cambria, Y. Li, F.Z. Xing, S. Poria, K. Kwok, SenticNet 6: Ensemble application of symbolic and subsymbolic AI for sentiment analysis, in: Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM), 2020, pp. 105–114.

[54] P.J. Stone, D.C. Dunphy, M.S. Smith, The General Inquirer: A Computer Approach to Content Analysis, MIT Press, 1966.

[55] P.C. Tetlock, M. Saar-Tsechansky, S. Macskassy, More than words: Quantifying language to measure firms' fundamentals, J. Finance 63 (3) (2008) 1437–1467.

[56] T. Loughran, B. McDonald, When is a liability not a liability? Textual analysis, dictionaries, and 10-ks, J. Finance 66 (1) (2011) 35–65.

[57] T. Loughran, B. McDonald, Textual analysis in finance, Annu. Rev. Financ. Econ. 12 (2020) 357–375.

[58] R. Correa, K. Garud, J.M. Londono, N. Mislang, Sentiment in central banks' financial stability reports, Rev. Finance 25 (1) (2021) 85–120.

[59] L. Barbaglia, S. Consoli, S. Manzan, Monitoring the business cycle with fine-grained, aspect-based sentiment extraction from news, in: Proceedings of MIDAS 2019: Mining Data for Financial Applications, in: Lecture Notes in Computer Science, vol. 11985, 2020, pp. 101–106.

[60] J.L. Fleiss, B. Levin, M.C. Paik, Statistical Methods for Rates and Proportions, John Wiley & Sons, 2003.

[61] J.L. Fleiss, C.M. Nee, J.R. Landis, Large sample variance of kappa in the case of different sets of raters, Psychol. Bull. 86 (5) (1979) 974–977.

[62] M. Friedman, A comparison of alternative tests of significance for the problem of m rankings, Ann. Math. Stat. 11 (1940) 86–92.

[63] G.W. Corder, D.I. Foreman, Nonparametric Statistics for Non-Statisticians: A Step-By-Step Approach, John Wiley & Sons, 2011, pp. 1–536.

[64] P.B. Nemenyi, Distribution-Free Multiple Comparisons (Ph.D. thesis), Princeton University, New Jersey, 1963.

[65] I. Brown, C. Mues, An experimental comparison of classification algorithms for imbalanced credit scoring data sets, Expert Syst. Appl. 39 (3) (2012) 3446–3453.

[66] G. Madjarov, D. Kocev, D. Gjorgjevikj, S. Džeroski, An extensive experimental comparison of methods for multi-label learning, Pattern Recognit. 45 (9) (2012) 3084–3104.

[67] J. Demšar, Statistical comparison of classifiers over multiple data sets, J. Mach. Learn. Res. 7 (2006) 1–30.

[68] M.T. Armesto, R. Hernández-Murillo, M.T. Owyang, J. Piger, Measuring the information content of the beige book: A mixed data sampling approach, J. Money Credit Bank. 41 (1) (2009) 35–55.

[69] S. Gerlach, Interest rate setting by the ECB, 1999–2006: Words and deeds, Int. J. Central Bank. 3 (3) (2007) 1–46.